

MUCKLE FLUGGA, United Kingdom

WMO: 030010

Lat: 60.855N

Lon: 0.886W

Elev: 174

StdP: 14.60

Time Zone: 0.00 (EUW)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.6	33.4	23.2	17.6	35.9	24.9	19.1	35.9	63.2	43.1	59.6	43.2	21.5	0	0.986

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	61.7	56.9	59.7	55.7	58.0	54.5	57.7	60.7	56.3	58.8	55.2	57.2	13.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.2	67.8	58.7	54.9	64.6	57.2	53.9	62.2	56.2	25.0	61.1	24.1	59.2	23.4	57.4	62.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 59.1	51.7	46.5	DB	28.7	66.3	2.5	2.8	26.9	68.4	25.5	70.0	24.1	71.6	22.3	73.7	(4)
(5)			WB	26.9	59.8	2.2	1.8	25.3	61.1	24.1	62.1	22.9	63.1	21.3	64.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.4	41.1	40.0	40.9	43.0	47.2	50.3	54.0	54.0	51.1	47.5	44.7	42.2
	DBStd	5.83	3.48	3.39	3.65	3.43	3.64	3.30	2.52	2.28	2.82	3.24	3.02	3.22
	HDD50	1695	275	281	281	211	104	34	1	0	19	87	160	242
	HDD65	6802	740	701	746	661	553	442	342	340	417	543	610	707
	CDD50	368	0	0	0	0	16	42	124	125	52	9	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	20.7	27.4	24.3	25.4	19.2	16.9	13.8	14.1	15.3	20.1	22.4	23.9	25.3
Precipitation	PrecAvg	47.6	5.4	3.7	4.6	3.0	2.2	2.3	2.3	3.0	4.5	5.2	5.7	5.6
	PrecMax	61.3	9.0	7.8	8.3	5.6	4.4	4.7	4.7	5.4	8.8	10.8	10.4	10.9
	PrecMin	38.4	1.7	1.1	1.2	0.2	0.6	0.2	0.9	0.6	1.1	2.4	1.8	3.1
	PrecStd	5.6	1.9	1.7	1.5	1.2	1.0	1.0	0.9	1.2	1.7	1.8	1.8	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.5	49.7	50.4	53.0	63.4	63.9	64.6	64.0	60.1	54.4	51.8	49.9
		MCWB	47.1	46.9	47.1	48.5	56.9	57.2	58.4	60.1	56.0	51.5	50.6	48.3
	2%	DB	48.4	47.6	48.5	50.3	59.1	60.0	61.6	60.2	57.4	53.0	50.4	48.6
		MCWB	46.9	45.6	46.0	46.8	53.5	54.7	56.9	57.0	54.1	50.7	49.0	46.9
	5%	DB	47.1	45.9	47.0	48.7	54.0	57.2	59.5	58.6	55.9	52.2	49.7	47.8
		MCWB	45.5	43.7	45.0	46.0	50.0	52.9	55.7	55.6	53.2	50.3	48.2	46.0
	10%	DB	45.9	44.9	45.8	47.5	51.4	54.8	57.7	57.2	54.7	51.5	48.9	47.0
		MCWB	44.0	42.8	43.8	45.0	48.1	51.7	54.4	54.5	52.3	49.7	47.2	45.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.3	47.4	47.9	49.2	57.4	57.5	59.4	60.4	56.6	52.5	50.8	48.8
		MCDB	48.8	48.5	49.4	52.1	63.7	63.0	63.0	62.5	58.4	53.7	51.2	49.4
	2%	WB	47.1	46.2	46.5	47.7	54.0	55.1	57.8	57.6	55.1	51.6	49.7	47.7
		MCDB	47.9	47.5	47.8	49.3	58.5	59.0	60.8	59.5	56.9	52.4	50.2	48.5
	5%	WB	45.8	44.3	45.5	46.7	50.5	53.6	56.3	56.2	53.8	50.9	48.6	46.5
		MCDB	47.0	45.5	46.8	48.2	53.3	56.4	58.7	57.8	55.1	51.9	49.5	47.7
	10%	WB	44.2	43.1	44.1	45.6	48.6	52.2	54.9	55.1	52.9	50.1	47.7	45.3
		MCDB	45.7	44.7	45.6	47.2	50.5	54.4	57.0	56.6	54.3	51.4	48.8	46.9
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.4	4.7	5.2	5.8	5.4	5.4	4.6	4.4	3.5	3.6	4.3
		MCDBR	5.4	5.6	5.6	7.0	9.7	9.4	8.4	6.7	5.9	4.0	4.3	5.2
		MCWBR	6.4	6.1	5.2	5.6	6.8	6.1	5.3	4.8	5.4	4.2	4.9	6.5
	5% WB	MCDBR	5.3	5.8	5.2	6.5	8.9	8.0	7.7	6.0	5.2	4.0	4.3	5.5
		MCWBR	6.5	6.5	5.5	5.7	6.7	5.6	5.5	4.7	5.4	4.6	5.2	7.1
Clear-Sky Solar Irradiance	taub	0.330	0.352	0.366	0.379	0.370	0.363	0.377	0.364	0.361	0.344	0.321	0.398	
	taud	2.071	2.196	2.276	2.313	2.404	2.463	2.443	2.478	2.441	2.317	2.126	2.358	
	Ebn at Noon	115	192	236	258	271	275	267	262	239	196	118	74	
	Edh at Noon	14	23	29	33	33	31	31	28	25	20	13	10	
All-Sky Solar Radiation	RadAvg	66	221	562	1060	1482	1532	1381	1127	679	335	98	41	
	RadStd	6	24	44	62	103	167	121	107	62	35	10	2	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	N/A	+1.17	+1.12	N/A	N/A	N/A	N/A	-184	-241	N/A	N/A

Nomenclature: See separate page